

Industry Problem Solving Task: Computer Vision System Selection

Technical Evaluation & Feasibility Assessment

Name:Shalini Douglas|Reg.No:192321019

1. Autonomous Vehicle Vision System Selection

System Constraints:

- Accuracy requirement: $\geq 90\%$ (If $< 90\%$ $\rightarrow$ system fails)
- Latency constraint: $\leq 70\text{ ms}$ (If $> 70\text{ ms}$ $\rightarrow$ unacceptable)
- Hardware support: Moderate computational resources

Method	Accuracy	Latency	Computation	Constraint Evaluation
Viola-Jones	82%	30 ms	Low	Fails (Accuracy $82\% < 90\%$)
YOLO	93%	60 ms	Moderate	Passes All Constraints
Deep Learning (Heavy)	97%	90 ms	High	Fails (Latency $90\text{ms} > 70\text{ms}$ & High Compute)

Decision & Justification:

YOLO is the optimal and only viable choice for the autonomous vehicle vision system.

- Accuracy: 93% exceeds the mandatory 90% threshold, whereas Viola-Jones fails at 82%.
- Latency: 60 ms meets the strict real-time requirement of $\leq 70\text{ ms}$, whereas Deep Learning fails at 90 ms.

- **Computation:** Moderate resource consumption aligns perfectly with the vehicle's onboard processing hardware.

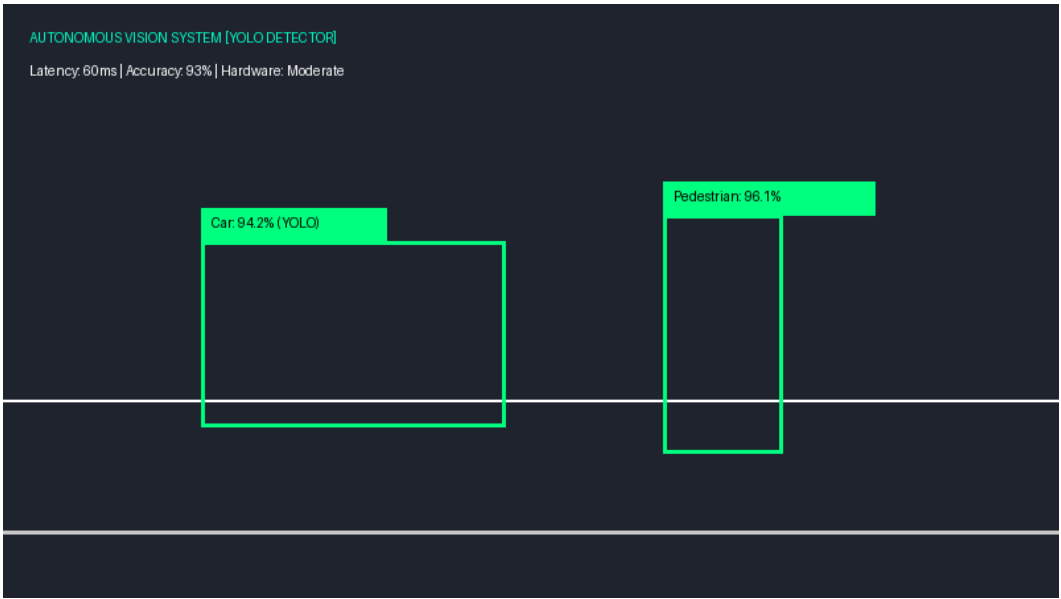


Figure 1: Example real-time autonomous vehicle object detection using YOLO.

2. Smart Factory Product Inspection Selection

System Constraints:

- Accuracy requirement: $\geq 95\%$ (If $< 95\%$ $\rightarrow$ defects may be missed)
- Processing time constraint: $\leq 50\text{ ms}$ (If $> 50\text{ ms}$ $\rightarrow$ production line is affected)
- Hardware support: Moderate computational resources

Method	Accuracy	Processing Time	Computation	Constraint Evaluation
Traditional Image Analysis	91%	25 ms	Low	Fails (Accuracy $91\% < 95\%$)
YOLO	96%	45 ms	Moderate	Passes All Constraints

Deep Learning (Heavy)	98%	75 ms	High	Fails (Time 75ms > 50ms & High Compute)
--------------------------	-----	-------	------	---

Decision & Justification:

YOLO is the only method that satisfies all requirements for the production line inspection system.

- Accuracy: 96% meets the stringent quality standard of $\geq 95\%$, preventing defective items from being missed (Traditional Image Analysis fails at 91%).
- Processing Time: 45 ms fits comfortably within the 50 ms cycle time limit to maintain production speed (Heavy Deep Learning fails at 75 ms).
- Computation: Moderate hardware load perfectly matches the smart factory deployment constraints.

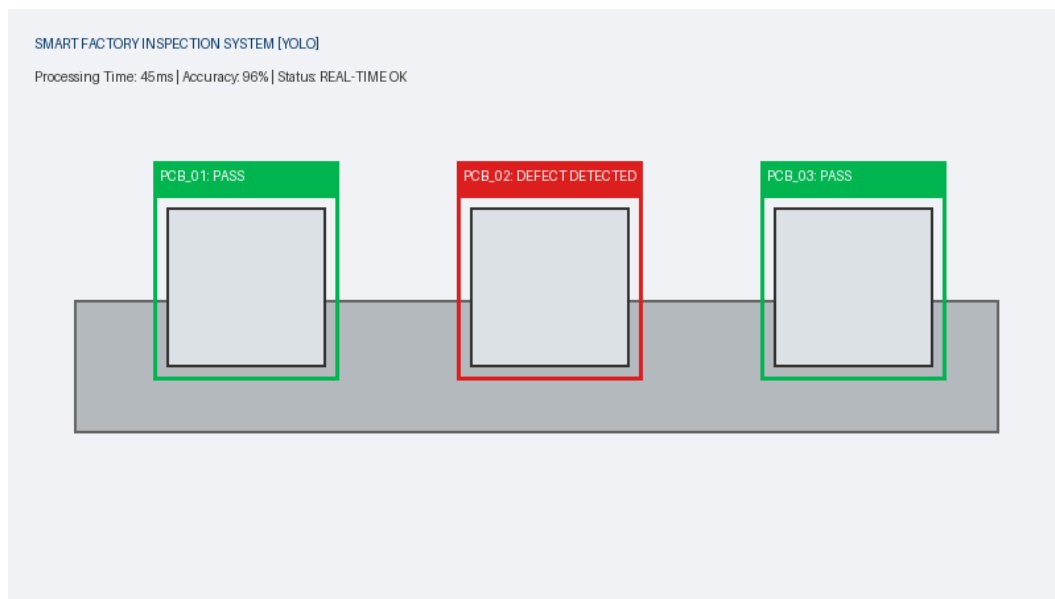


Figure 2: Example industrial conveyor belt defect detection visual model using YOLO.